



Foreword

"The past is never dead. It's not even past."
William Faulkner, *Requiem for a Nun*, 1950

Modern science carries multiple traces of its historical origins: we encounter its past every day. Even the most sophisticated clocks mark off time in sixties, a survival from Babylonian numbering systems used many thousands of years ago. Scientific heroes are celebrated in units of measurement—Volts, Curies, Richters—and in parts of our body, such as the Eustachian tubes in our ears. Discarded scientific theories live on in language: “melancholic” and “sanguine” originated in ancient Greek medicine, while “mesmerizing” refers to an 18th-century French therapy based on magnets. Plants and animals still bear the Latin names of Carl Linnaeus’s classification system, introduced in Sweden long before Charles Darwin’s evolutionary theories made sense of life’s complicated variety—and rainbows have seven colors because Isaac Newton believed they should follow the mathematics of musical scales worked out by Pythagoras.

Technological science now permeates society, inseparable from political, commercial, military, and industrial projects, yet the word “scientist” was invented only in 1833. Despite that apparently late start, science has ancient roots. Long before universities and laboratories were created, stargazers studied the heavens to calculate the dates of

religious festivals, while scholars attached to mosques and monasteries deciphered God’s designs by interpreting the natural world. Uneducated men and women were building up the practical expertise that later provided the foundation of scientific disciplines—how to distil medicines from herbs, smelt ores to produce metals, navigate by the stars, detect the signs of bad weather, mix chemicals to make soap.

From the earliest attempts to make fires, pots, and tools, people have always experimented to find out how the world works and how they can make their lives more comfortable. These twin goals of scientific research were spelled out in the early 17th century by philosopher Francis Bacon. “Knowledge is power,” he declared, and the rate of change accelerated as governments increasingly recognized the advantages to be gained from investment in scientific projects. Expanding exponentially, technological science rapidly came to dominate the world, uniting it in an international web of instantaneous electronic communication.

Science has uncovered many of nature’s secrets, but it has also unleashed some genies—atomic energy, global warming, genetic modification—that may ultimately destroy us. As citizens of a scientific global community, we need to understand the past in order to control our own future.

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Extremophile habitat

Vivid colors in the Grand Prismatic Spring in Yellowstone National Park, US, result from a film of pigmented bacteria around the edge of the hot spring. Different species of microbes flourish in specific temperatures and contain pigments suited to their environments.